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[5151]-108

F.E. EXAMINATION, 2017

ENGINEERING MATHEMATICS-II

(2015 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt *four* questions : Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of electronic non-programmable calculator is allowed.

(v) Assume suitable data if necessary.

1. (A) Solve the following differential equations : [8]

(i) $x^4 \frac{dy}{dx} + x^3 y = \sec(xy)$

(ii) $\frac{dy}{dx} = \frac{1 + y^2 + 3x^2 y}{1 - 2xy - x^3}$

(B) A body starts moving from rest is opposed by a force per unit mass of value cx and resistance per unit mass of a value bv^2 , where x and v are the displacement and velocity of the particle at that instant. Show that the velocity of the particle is given by :

$$v^2 = \frac{c}{2b^2} (1 - e^{-2bx}) - \frac{cx}{b} \quad [4]$$

P.T.O.

Or

2. (A) Solve : [4]

$$\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y.$$

- (B) Solve the following : [8]

(i) Water at temperature 100°C cools in 10-minutes to 88°C in a room of temperature 25°C . Find the temperature of water after 20 minutes.

(ii) A resistance of 100Ω , an inductance of 0.5 henry are connected in a series with battery of 20 volts. Find the current in a circuit as a function of time t .

3. (a) Find Fourier series to represent the function $f(x) = x$ in $-\pi < x < \pi$ and $f(x) = f(x + 2\pi)$. [5]

- (b) Evaluate :

$$\int_0^\infty \sqrt{y} \cdot e^{-\sqrt{y}} dy \quad [3]$$

- (c) Trace the curve (any one) : [4]

(i) $y^2 (x^2 - 1) = x$

(ii) $r = a(1 + \cos \theta)$.

Or

4. (a) If :

$$I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta \, d\theta,$$

prove that :

$$I_n = \frac{1}{n-1} - I_{n-2}. \quad [4]$$

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(b) Prove that :

$$\int_0^1 \frac{x^a - x^b}{\log x} dx = \log \left(\frac{a+1}{b+1} \right), \quad a > 0, b > 0 \quad [4]$$

(c) Find the length of the arc of cycloid [4]

$x = a (\theta + \sin \theta)$, $y = a (1 - \cos \theta)$
between two consecutive cusps.

5. (a) Find the centre and radius of the circle which is an intersection of the sphere $x^2 + y^2 + z^2 - 2y - 4z - 11 = 0$ by the plane $x + 2y + 2z = 15$. [5]

(b) Find the equation of the right circular cone which passes through the point $(1, 1, 2)$ & has its axis along the line $6x = -3y = 4z$ and vertex at $(0, 0, 0)$. [4]

(c) Find the equation of a right circular cylinder of radius 2 whose axis passes through $(1, 2, 3)$ and has direction cosines proportional to $2, -3, 6$. [4]

Or

6. (a) Show that the plane $4x - 3y + 6z - 35 = 0$ is tangential to the sphere $x^2 + y^2 + z^2 - y - 2z - 14 = 0$. [5]

(b) Find the equation of a right circular cone whose vertex is at $(1, 2, 3)$ and axis has direction ratios $(2, -1, 4)$ and semivertical angle 60° . [4]

(c) Find the equation of the right circular cylinder of radius 3 whose axis is the line

$$\frac{x-1}{2} = \frac{y-3}{2} = \frac{z-5}{-1}. \quad [4]$$

7. Attempt any *two* of the following :

(a) Evaluate

$$\iint \frac{x^2 y^2 dx dy}{x^2 + y^2}$$

where R is annulus between $x^2 + y^2 = 4$, $x^2 + y^2 = 9$. [6]

(b) Evaluate

$$\iiint (x^2 y^2 + y^2 z^2 + z^2 x^2) dx dy dz$$

throughout the volume of sphere $x^2 + y^2 + z^2 = a^2$. [7]

(c) Find the moment of inertia of one loop of lemniscate $r^2 = a^2 \cos 2\theta$ about initial line. [6]

Or

8. Attempt any *two* of the following :

(a) Find the total area included between the two cardioids

$$r = a (1 + \cos \theta) \text{ and } r = a(1 - \cos \theta). \quad [6]$$

(b) Find the volume cut-off from the paraboloid $x^2 + \frac{y^2}{4} + z = 1$ by

the plane $z = 0$. [7]

(c) Find the C.G. of an area of the cardioid :

$$r = a (1 + \cos \theta). \quad [6]$$